Bow Valley College

Course Outline

DATA3403

Designing Data Analysis Projects
Winter 2023 - Spring 2025

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DATA3403 Designing Data Analysis Projects

COURSE DESCRIPTION

Effective data analysis requires the integration of business understanding and data understanding. In this course, learners synthesize information about common business processes, workflows, and management strategies across a range of sectors (such as sales, marketing, accounting, quality improvement, product/service delivery, product development, and human resources), and their associated data needs. Learners then apply this knowledge to real-world business contexts to identify and define business goals and design appropriate data projects

REQUISITES	None
EQUIVALENTS	None
CREDITS	3
HOURS	45
ELIGIBLE FOR PLAR	No
ZERO TEXTBOOK COST	

COURSE COMPETENCY

COMPETENCY TITLE

Data Management and Analytics – Designing Data Projects

COMPETENCY STATEMENT

Data Management and Analytics professionals apply business analysis and project design skills to design data projects that will meet stakeholder needs and help to optimize business operations.

COMPETENCY DESCRIPTION

Project design is a critical phase in the data analysis process as it sets the foundation for high-quality and meaningful data insights. Informative data analysis projects integrate business needs and data opportunities that meet the needs of stakeholders.Well-designed data projects ensure that a business problem or question can be answered through data, that the right analysis approach is applied, and that deadlines and critical milestones can be met. Before embarking on data acquisition, preparation and analysis, data analysts and scientists apply their understanding of business processes, strategies, and data to identify and define business goals, and plan data projects that meet time, budget, and functional objectives.With the emergence of data-driven solutions, data project design represents a critical job function across many high-value sectors and industries.

COURSE LEARNING OUTCOMES

Bow Valley College is committed to ensuring our graduates can demonstrate their abilities in key areas that will make them effective citizens and encourage their development as lifelong learners. In addition to the discipline-specific skills that learners acquire in their programs, the College has identified ten learning outcomes.

College-Wide Outcomes:

1. Communication
2. Thinking Skills
3. Numeracy and Financial Literacy
4. Working with Others
5. Digital Literacy
6. Positive Attitudes and Behaviours
7. Continuous Learning
8. Health and Wellness Awareness
9. Citizenship and Intercultural Competence
10. Environmental Sustainability

#	COURSE LEARNING OUTCOME(S)	COLLEGE WIDE OUTCOMES SUPPORTED
1	Determine the suitability of a data-driven approach to solve a business problem by demonstrating problem-solving and critical-thinking skills.	1, 2, 3, 5, 6
2	Create a project plan for a data analysis project that meets business needs by demonstrating decision-making and communication skills.	1, 2, 3, 5, 6

LEARNING PATHWAY

**The time it takes learners to demonstrate competencies will vary. An example of a suggested schedule for learning and development is shown below. Learners will need to plan out their assessment attempts within their course. For additional information, please consult the Course Offering Information in Brightspace.*

WEEK/HOURS	LEARNING AND DEVELOPMENT PLAN
Week 1	Review Performance Demonstrations, Competency Assessment, and learning pathway
Week 2	Create Success Plan - Complete Performance Demonstration
Week 3	Determine business objectives
Week 4	Determine business objectives
Week 5	Verify feasibility of project with respect to business needs and available resources
Week 6	Determine data analysis goals
Week 7	Determine data analysis goals
Week 8	Determine desired data features
Week 9	Reading week
Week 10	Determine desired data features
Week 11	Propose data analysis methodologies and tools
Week 12	Propose data analysis methodologies and tools
Week 13	Create a personal project schedule
Week 14	Performance Demonstration
Week 15	Competency Assessment

ASSESSMENT

This course follows an assessment-first approach, in which learners will be assessed, and receive structured feedback, and a personalized learning plan. Learners will also receive differentiated support from an instructor based on their individual needs.

Learners will have a variety of ways to demonstrate they have met the required competency through the demonstration of learning outcomes and criteria as laid out in the rubric. Learners will have multiple (but not unlimited) attempts to prove competency. It is suggested that learners plan out their assessment attempts within their course.

Learners will have flexibility in how they satisfy course learning outcomes while still adhering to the criteria found in the rubric and the Course Offering information. Please refer to the Course Offering Information and the rubric in Brightspace for additional information.

COURSE LEARNING OUTCOMES	ASSESSMENT
1	Performance Demonstration
2	Performance Demonstration

PERFORMANCE STANDARDS

Successful demonstration of all learning outcomes is required to pass this course. This course may be part of a competency or larger program of study.

Please consult with the program area or the Bow Valley College website for completion requirements.

GRADING SCHEME

Symbol	**Description**	**Grade Point Value**
MAST	Mastery	4.00
COMP	Competent	3.00
DEVL	Developing	N/A
F	Fail	0.00
W	Withdraw	N/A

REQUIRED LEARNING RESOURCES

Additional learning resources may be found in the Course Offering Information or in Brightspace.

ADDITIONAL INFORMATION

The Designing Data Projects competency makes up part of the Data Management and Analytics job profile, along with:

- Acquiring and Wrangling Data
- Performing Data Analysis
- Visualizing and Presenting Insights

Additional information may be found in the Course Offering Information or in Brightspace.

ACADEMIC ACCOMMODATIONS

Learners with a disability (learning, physical, and/or mental health) may qualify for academic and exam accommodations. For more information, or to apply for accommodations, learners should make an appointment with Accessibility Services in the Learner Success Services (LSS) Department. Accessibility Services can also assist learners who may be struggling with learning but do not have a formal diagnosis. To make an appointment visit LSS on the first floor of the south campus or call 403-410-1440. It is the learner's responsibility to contact Accessibility Services and request academic accommodations. For more information, please visit our website at http://www.bowvalleycollege.ca/accessibility.

INSTITUTIONAL POLICIES

Bow Valley College is committed to the highest standards of academic integrity and honesty. Learners are urged to become familiar with and uphold the following policies: Academic Integrity (500-1-7), Learner Code of Conduct, Procedures and Guidelines (500-1-1), Learner Appeals (500-1-12), Attendance (500-1-10), Grading (500-1-6), Academic Continuance and Graduation (500-1-5), and Electronic Communications (300-2-13). Audio or video recording of lectures, labs, seminars, or any other teaching and learning environment by learners is allowed only with consent of the instructor as part of an approved accommodation plan. Recorded material is to be used solely for personal study and is not being used or distributed without prior written consent from the instructor.

Turnitin:

Students may be required to submit their course work to Turnitin, a third-party service provider engaged by BVC. Turnitin identifies plagiarism by checking databases of electronic books and articles, archived webpages, and previously submitted student papers. Students acknowledge that any course work or essays submitted to Turnitin will be included as source documents in the Turnitin.com reference database, where it will be used solely to detect plagiarism. The terms that apply to a student's use of Turnitin are described on Turnitin.com.

Online Exam Proctoring:

Examinations for this course may require proctoring through an online proctoring service. Online proctoring enables online exam taking within a controlled and monitored environment, thereby enhancing academic integrity. Online proctoring may occur through a variety of methods, including but not limited to:

a. live online proctoring where a remote invigilator authenticates identity and observes completion of an exam using specialized software and recordings;
b. automated proctoring where the exam session is recorded and AI (artificial intelligence) analyzed;
c. browser lockdown that limits access to other applications, websites, copying, printing, screen capture and other functions; or
d. a combination of both live/automated proctoring and browser lockdown.

Course instructors will review recordings, analyses, and data obtained through online proctoring for academic integrity infractions. It is the student's responsibility to meet the technical, software, location, and identity verification requirements necessary to enable online proctoring.

Further details of these policies are available in the Academic Calendar and on the Bow Valley College website, bowvalleycollege.ca.

Learners are encouraged to keep a copy of this course outline for future reference.

Collection of Personal Information:

This course, including your image and voice, may be recorded and made available to you and other students taking the course section. By attending the class(es) online or in person, you consent to the collection of your personal information. If you do not wish to be recorded, please contact your instructor before starting the course/class to discuss alternative arrangements.

You may use the recordings only for educational purposes and you must not copy, share, or use the recordings for any other purpose without the instructor's express permission.

Your personal information is collected in accordance with section 33(c) of the Freedom of Information and Protection of Privacy Act (Alberta) to deliver academic programming, support learner flexibility, promote universal design for learning principles, and for purposes consistent with the course activities and outcomes. If you have any questions about the collection, disclosure, use, or protection of this information, please contact the College's Access and Privacy Officer at foip@bowvalleycollege.ca.